ZIP 94158: Mission Bay, Dogpatch, and east Potrero Hill, San Francisco

The people most likely to die here this winter are the ones sleeping in vehicles and tents on the Dogpatch and Potrero streets, not the people in the new towers. The flooding is real but survivable. What kills is a cold, wet night with nowhere dry to go, a generator or heater run inside an RV, or using drugs alone during a storm. The single most important step is to get the city to open storm-night shelter on a written trigger and to stop clearing encampments while it is open.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
San Francisco County, CA (inferred; the city and county share one boundary)	None published for the US, California, or San Francisco	n/a here; global 6-month figure 239,000 ± 11,000	Atmospheric-river nights: cold, wet exposure for people living outside or in vehicles, carbon monoxide during outages, and sewer-backup flooding on the western edge of the ZIP

EXECUTIVE SUMMARY — AT A GLANCE



37.77°N 122.39°W, United States of America · Natural Earth

- **The risk.** Those most likely to die are sleeping in vehicles and tents on the Dogpatch and Potrero streets: a cold, wet night with nowhere dry to go, a generator or heater run inside an RV, or using drugs alone during a storm.
- **First action.** HSH, DEM, and Episcopal Community Services open storm-night shelter on a published trigger, with an east-side walk-up site and a written pause on encampment and vehicle enforcement while it is open.
- **The money.** $191k floor, $489k target, $1.26M ceiling; the floor starts with 100 surge beds for 10 nights and $20,000 of kits.
- **Do not.** Do not clear encampments or tow lived-in vehicles in the days around a storm.

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* Suggested formulation: check with a health worker before sharing.

13/13 recommendations checked. 1 confirmed, 11 corrected, 0 unverified, and 1 contraindicated (parametric household flood cover for this winter). Two further practices are flagged “do not do”. Exposure figure: Climate Impact Lab, September 2026, pre-peer-review, heat only. No figure exists at this resolution, and heat is not this place's winter hazard, so no number is borrowed. **The funding target fell 49% in verification**, from about $950k to $489k. The floor had been added up wrong, the city already funds storm-night beds on about ten nights a winter, and the kit budget was sized to the whole city rather than this ZIP.

Where this is. ZIP 94158 is a postal area, not a neighbourhood. It covers southern Mission Bay (Mission Creek south to about Mariposa Street, including UCSF Mission Bay, Benioff Children's Hospital, and Chase Center) plus parts of Dogpatch and east Potrero Hill. North Mission Bay is in 94107, and Islais Creek is outside it, in 94124. It splits across two supervisorial districts: Mission Bay is in District 6 and Dogpatch and Potrero are in District 10. Most statistics below are district-level or citywide, and are labelled as such. Mission Bay was built on raised fill, and the SFPUC 100-Year Storm map (2022 version) touches only the ZIP's western fringe, around Mission Creek, 7th, 16th, and Hooper Streets. The Mission Bay core and Dogpatch along Third Street fall outside it.

What the forecast says. NOAA's Climate Prediction Center (8 October 2026) has an El Niño Advisory in force and calls a strong-to-very-strong El Niño likely through January–March 2027. It gives the California coast 50–60% odds of a wetter-than-normal winter. That is likely, not certain: the very strong 2015–16 El Niño left San Francisco at about 98% of normal, while the city's two wettest days on record came in a weak El Niño (November 1994) and a La Niña (31 December 2022). Prepare for atmospheric rivers in any winter. This one makes them more likely.

WHAT THIS PLACE SHOULD DO RANKED

Actions are ranked by expected lives saved per unit of effort before the peak. The backing lenses are shown on each action, and the RACI chart below assigns each one. Nearly everything here strengthens a program the city already runs. That is a finding: the gaps are triggers, location,

* Suggested formulation: check with a health worker before sharing.

and enforcement, not missing programs.

1. Get storm-night shelter opened on a published trigger, with an east-side walk-up site and a written pause on encampment and vehicle enforcement while it is open

PARTIALLY CONFIRMED | EMERGENCY MGR · SKEPTIC · HISTORIAN · PHILANTHROPIST · FRONTLINE · PUBLIC HEALTH

INVENTORY: QUERIED, NO PRECEDENT

OWNER	BY WHEN	COST	PREVENTS
Department of Homelessness and Supportive Housing (HSH), with the Department of Emergency Management (DEM); Episcopal Community Services (ECS) runs the beds; Healthy Streets Operations Center (HSOC) for the enforcement pause	**Triggers and the east-side site agreed by 15 Nov; 2026–27 Interfaith Winter Shelter announced before Thanksgiving**	**Policy work; surge nights beyond the city's own at ~$126 per occupied bed-night (see funding)**	**Cold, wet exposure deaths among people living outside or in vehicles on storm nights**

HSH already runs a Cold & Wet Weather Protocol. In 2025–26 it opened Next Door (1001 Polk) and Sanctuary (201 8th St, an ECS site) for about ten nights, 31 December to 5 January and 15 to 18 February, and the street outreach team (SFHOT) handed out blankets and ponchos. Three things are missing. **The numeric triggers are not published**, and HSH opens beds at its discretion. **No site is east of 8th Street**, so a person in an RV on Cesar Chavez has to cross SoMa in the rain. **No rule pauses enforcement during an activation**: storm-time policy is to “focus on soiled materials”, and KQED documented tent seizures before storms in November 2024. In January 2023, 297 shelter beds were open, many reserved, against more than 4,000 people unsheltered. Ask HSH to add NWS flood, high-wind, and atmospheric-river (AR3 or higher) forecasts to its triggers, to put a Bayview, Dogpatch, or Mission Bay site on its activation list, and to have outreach carry same-day bed counts. Toronto's data show most cold injuries among unhoused people happen in moderate, wet weather rather than extreme cold, which argues for low triggers.

Corrections: this is a strengthening of an existing protocol, not a new program. The $126.25 rate is real (SF Controller, December 2024 assessment), but it is the FY23 average cost per occupied adult and youth bed-night across all programs, not the marginal cost of a surge night. The “437 of 501 large vehicles lived in” figure is HSOC data from May 2025, reported by Potrero View. Pier 68 storage for towed RVs

* Suggested formulation: check with a health worker before sharing.

was paused until at least 1 March 2026, with no vehicles towed there. The District 10 “+78% unsheltered” figure could not be found; the 2024 count shows a “sharp rise” in District 10 and citywide vehicle dwelling up 37%. No study was found showing that storm or cold shelter activation reduces deaths, so this action rests on plausibility and on the documented gap, not on an effect size.

2. Put RV-listed carbon monoxide alarms in lived-in vehicles and houseboats, and say “no generator, grill, or propane heater inside” in English, Spanish, and Chinese

PARTIALLY CONFIRMED | PUBLIC HEALTH · SKEPTIC · PHILANTHROPIST · FRONTLINE | INVENTORY: QUERIED, NO PRECEDENT

OWNER	BY WHEN	COST	PREVENTS
Outreach workers the vehicle dwellers already trust (not sweep crews), with SF Fire Department; building operators pass the message to households	**Before the first king tide window, 23 Nov**	**$5k–10k (assumed unit cost; no bulk quote found)**	**Carbon monoxide deaths in vehicles and homes during storms and outages**

Outages raise carbon monoxide poisoning sharply. In New York, 2017–2020, four hours without power during an outage affecting at least 20% of a community raised the odds of a child's CO emergency visit by 150% or more that day (Northrop et al., *Pediatrics* 2025, 917 cases). In January 2023 two men died at Sea Ranch with a generator running in a hallway; CO was suspected, though the coroner had not confirmed it when reported. California requires CO alarms in dwelling units (Health & Safety Code §17926), but no state law requiring them in RVs was found. Buy alarms listed to UL 2034 for RV use. Hand them out with the frontline lens's warning in mind: people living in RVs read any official-looking flyer as a tow notice, so the alarm has to come from someone who has never been with the sweep crews.*

* Suggested formulation: check with a health worker before sharing.

Corrections: the *Pediatrics* population is under-18s, not under-fives. The ~200 vehicles and ~$50 per alarm figures are unverified estimates. Evidence for the mechanism is strong; evidence for alarm distribution in RVs specifically is absent. This area is family-heavy (over a third of residents at 1180 Fourth St are under 18), which strengthens the household half of this action.

3. Keep a building-level outage roster in Mission Bay's supportive and affordable housing, and feed it into the city's existing outage wellness checks

PARTIALLY CONFIRMED | FRONTLINE · PUBLIC HEALTH · PHILANTHROPIST | INVENTORY: QUERIED, NO PRECEDENT

OWNER	BY WHEN	COST	PREVENTS
Building operators (e.g. Mercy Housing at 1180 Fourth St) and floor captains; Department of Disability and Aging Services (DAS) runs the checks; SF Fire Department's NERT volunteers support	**Rosters by 1 Dec**	**Staff and volunteer time**	**Deaths among people who depend on elevators, oxygen, or powered devices when the power is out**

A live HHS emPOWER query on 9 October 2026 shows 907 Medicare beneficiaries in 94158, of whom 30 use power-dependent equipment. That number is small enough to know by name. DAS already ran wellness checks during the 20 December 2025 outage, when a substation fire cut power to about 130,000 customers, so the roster should feed DAS rather than duplicate it. Each check covers three things: warmth, a CO source, and power for devices or oxygen. The 2003 New York blackout raised non-accidental deaths by 25% and accidental deaths by 122% (Anderson & Bell, *Epidemiology* 2012). On 4 February 2024, PG&E had 1.1 million customers out at peak across its territory.*

* Suggested formulation: check with a health worker before sharing.

Corrections: 1180 Fourth St is *family* housing (150 units, 50 for formerly homeless families), not senior housing. ECS was its services partner at opening; the current provider was not confirmed. The emPOWER sub-counts for oxygen, dialysis, and home health (11 each) are probably a privacy floor, not real counts. PG&E's Medical Baseline gives extra notice and a door knock, but **not priority restoration**; do not tell people it does. NERT's size is given as 24,700 and as 29,700 in different SFFD documents. No study measures how much welfare checks reduce outage deaths; the blackout study measures the harm, not the fix.

4. On storm nights, add naloxone and “don't use alone” outreach at the shelters and at RV and tent clusters

PARTIALLY CONFIRMED · ADD-ON | PUBLIC HEALTH | INVENTORY: QUERIED, NO PRECEDENT

OWNER	BY WHEN	COST	PREVENTS
SF Department of Public Health (DPH): DOPE Project and Naloxone Clearinghouse, through the same outreach teams as action 1	**Built into the first activation**	**Within existing DPH supply**	**Overdose deaths, the leading cause of death among unhoused San Franciscans**

Overdose, not cold, is what kills most unhoused people in San Francisco. In the year from March 2020, 331 unhoused people died, against 128–147 a year in 2016–19, and overdose caused 178 of the 216 deaths with a known cause (*JAMA Network Open* 2022). Citywide overdose deaths were 810 in 2023, 633 in 2024, and 624 in 2025 (preliminary); January–August 2026 is about 24% below the same months of 2025 (Office of the Chief Medical Examiner). Overdose education and naloxone distribution cut overdose death rates in Massachusetts towns (rate ratio 0.73 at low uptake, 0.54 at high uptake; Walley et al., *BMJ* 2013).

* Suggested formulation: check with a health worker before sharing.

Corrections: **no study links rain or storms to overdose deaths.** The only weather study found is on cold (Rhode Island and Connecticut: a 7-day average of 0°C carried about 1.3 times the odds of fatal overdose compared with 11°C, *Epidemiology* 2019), and San Francisco rarely gets near 0°C. The Walley effect size comes from the pre-fentanyl era and should not be carried over to today's San Francisco. DPH already distributes naloxone at scale (about 100,000 doses in January–September 2023), so this is a storm-night targeting of an existing program, and it is mostly a year-round measure rather than an El Niño one.

5. Get the Department of Emergency Management's current activation trigger in writing, and press for automatic activation on an NWS flood watch

PARTIALLY CONFIRMED | EMERGENCY MGR | INVENTORY: QUERIED, NO PRECEDENT

OWNER	BY WHEN	COST	PREVENTS
DEM; District 6 and District 10 supervisors' offices to ask the question	**15 Nov**	**Staff time**	**A repeat of 31 Dec 2022, when the city opened its Emergency Operations Centre while the storm was already under way**

On 31 December 2022, 5.46 inches fell, the city's second-wettest day on record. The winter storm plan treated less than an inch as no threat, and the Emergency Operations Centre opened late in the morning, although the NWS had issued a flood watch days earlier. Residents filed 117 home and 17 business damage reports, and FEMA declined individual grants. Activate early and stand down if the storm misses. Separately, the city should check that its Cal OES agent resolution (form OES-FPD-130, valid for three years) is current, so that the state's 75% reimbursement of public response costs under the California Disaster Assistance Act can be claimed from day one.

* Suggested formulation: check with a health worker before sharing.

Corrections: no published after-action report and no revised trigger were found, so DEM may already have changed its rule. Ask before proposing a new one. The CDAA reimbursement is confirmed, but it only reaches public agencies, not households.

6. Deliver flood barriers and sandbags to addresses with a backup history and to affordable buildings, instead of relying on the pickup yard

PARTIALLY CONFIRMED | SKEPTIC · HISTORIAN (AS CORRECTED) · EMERGENCY MGR | INVENTORY: QUERIED, NO PRECEDENT

OWNER	BY WHEN	COST	PREVENTS
SF Public Works with SFPUC	Delivery list by 1 Dec	Not quantified	Sewer-backup and street flooding into ground-floor homes of people without cars

Today Public Works sandbags are pickup-only: ten bags per address, from the Marin and Kansas yard, Monday to Saturday, 8am to 2pm. That design excludes renters without cars and residents of affordable and supportive housing. In the 2015–16 season, Public Works and SFPUC ran neighbourhood “Sandbag Saturday” pop-ups and handed out more than 20,000 bags between them. Use that model, plus delivery to known backup addresses on the ZIP's western fringe.

Corrections: the report that the city repeatedly ran out of sandbags on 31 December 2022 is in the *Chronicle*, behind a paywall, and was not read in the original. The missing Mission barriers are confirmed for 17th and Folsom, which is outside this ZIP. This resolves the panel's sandbag disagreement: pop-ups and delivery, not a single yard.

7. Recruit owners who have already had a sewer backflow into the SFPUC Floodwater Grant now, for installation after this winter if not before

PARTIALLY CONFIRMED · DEMOTED | ECONOMIST · HISTORIAN | INVENTORY: QUERIED, NO PRECEDENT

OWNER	BY WHEN	COST	PREVENTS
SFPUC (payer); building owners apply; neighbourhood associations spread the word	Interest forms in by 15 Nov	Paid by SFPUC: up to $100k per property, 100% of eligible costs	Repeat sewer backups into ground floors

* Suggested formulation: check with a health worker before sharing.

This is the only money here that moves before a flood rather than after one. SFPUC reimburses up to 100% of eligible costs, to $100,000 per property, for measures such as backwater valves and flood barriers. Hardship owner-occupiers and small businesses or nonprofits with 50 or fewer staff can get an advance of the lesser of 10% of the project cost or $1,000.

Corrections and demotion: eligibility requires a **prior storm-related flood or backflow event**, not a location on the 100-Year Storm map, so target by history. The grant agreement must be signed before work starts, and the steps (interest form, site visit, two quotes and drawings, permit, then a six-month build window) make installation before the peak unlikely. The Wawona precedent one lens cited was a $29M capital sewer project, not this grant. No causal study of backwater valves was found; their value is engineering consensus.

8. Before the first winter bill arrives, build a queue of households for PG&E's REACH and Match My Payment help

PARTIALLY CONFIRMED | ECONOMIST | INVENTORY: QUERIED, NO PRECEDENT

OWNER	BY WHEN	COST	PREVENTS
PG&E (payer); building operators and community groups help households apply	**Now; the funds are first come, first served**	**Paid by PG&E's $50M 2026 pledge**	**Households going into an outage season already disconnected or in arrears**

REACH pays up to $800 to households at or below 200% of the federal poverty level that have a past-due balance and a 15-day or 48-hour disconnection notice. It can be used once a calendar year, and households on PG&E's Arrearage Management Plan are excluded. Match My Payment matches up to $1,000 for households up to 400% of the poverty level, with no disconnection notice needed. REACH recipients prequalify for it, so the combined maximum is $1,800. CARE and FERA bill discounts are existing enrolment.

* Suggested formulation: check with a health worker before sharing.

Corrections: Match My Payment's income limit is 400%, not 200%, of the poverty level. **Do not cite a winter shutoff moratorium**: the CPUC rule bars disconnection only when the 72-hour forecast is below 32°F or above 100°F, which San Francisco rarely reaches. A draft resolution (E-5468, 2026) would revise the thresholds, but its adoption was not confirmed.

9. Set up a buddy roster for the Mission Creek houseboats around the three highest tides of the winter, mainly for safe access and to avoid sewage contact

PARTIALLY CONFIRMED · LOW MORTALITY | FRONTLINE · EMERGENCY MGR · SKEPTIC · HISTORIAN | INVENTORY: QUERIED, NO PRECEDENT

OWNER	BY WHEN	COST	PREVENTS
The Mission Creek houseboat community, with the Port of San Francisco	Before 23 Nov	Negligible	Injury on gangways, loss of utilities, and illness from sewage-overflow contact

About 20 houseboats sit on Mission Creek, beside outfalls that release combined sewage in heavy storms (Baykeeper). NOAA predicts the winter's highest tides at the San Francisco station (9414290) on 23–26 November (peak 7.03 ft above MLLW), 22–25 December (peak 7.23 ft, 24 December at 10:37am), and 20–23 January (peak 7.11 ft, 21 January at 9:39am). Pair each boat with a land-side contact, check in on mooring, power, and access during those windows and during any storm that falls on them, and post water-contact warnings after outfall releases.

Corrections: the homes float, and rose about 3 ft in the March 2011 tsunami; no houseboat flood event was found. The risks are access, utilities, and sewage, so the expected mortality benefit is low. The “~35 boats” and “moved from Islais Creek” claims were not confirmed. The California King Tides Project had not yet posted 2026–27 dates.

Checked and set aside here: priority inspection of street trees at bus stops (Emergency mgr). Two people were killed by falling trees in San Francisco on 21 March 2023, on the 1000 block of Post Street and at a Lincoln Way bus stop, but neither was in this ZIP. Mission Bay's street trees are mostly young plantings, and Third Street's T-line uses platforms, not bus stops. Public Works already owns all street trees under Proposition E (2016), with hazards reported through 311. Partially confirmed; low expected yield in 94158.

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WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	HSH	DEM	ECS	DPH	DAS	SF Fire / NERT	Public Works	SFPUC	Port	PG&E	Building operators	Trusted outreach groups	Supervisors D6 / D10
1. Storm-night shelter trigger; east-side site; enforcement pause	**A/R**	**R**	**R**	**C**			**C**					**R**	**C**
2. CO alarms; three-language message				**C**		**A/R**			**I**		**R**	**R**	
3. Building outage roster into DAS checks		**C**		**C**	**A/R**	**R**				**I**	**R**		
4. Storm-night naloxone	**C**		**R**	**A/R**								**R**	
5. EOC trigger in writing; CDAA resolution	**I**	**A/R**				**C**	**C**	**C**					**R**
6. Barrier and sandbag delivery		**C**					**A/R**	**R**			**C**		**I**
7. Floodwater Grant sign-ups								**A**			**R**	**R**	**I**
8. PG&E REACH and Match My Payment queue					**C**					**A**	**R**	**R**	
9. Houseboat buddy roster		**I**		**C**				**C**	**A**			**R**	

What the chart shows. **The highest-ranked action has no accountable owner who wants it.** HSH is accountable for action 1, but the enforcement pause sits with HSOC and Public Works, and the record shows clearings continuing during storms. The supervisors' offices are the lever for that row. **Action 2 has no natural owner for vehicle dwellers**; the Fire Department is listed because CO is its remit, but the alarms only land if trusted outreach groups carry them.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs. No placement carries a published cost per death averted; amounts are unit costs times assumed quantities, and the assumptions are shown.

FLOOR	TARGET	CEILING	LARGEST UNCERTAINTY
$191k	**$489k**	**$1.26M**	**How many storm nights there will be, and whether HSH funds its own surge for 2026–27**

Floor: surge beds 100 × 10 nights × $126.25 = $126,250 + kits $20,000 + CO alarms $5,000 + SF Foundation pool $40,000 = **$191,250**. **Target:** 150 × 20 × $126.25 = $378,750 + $40,000 + $10,000 + $60,000 = **$488,750**. **Ceiling:** 200 × 40 × $126.25 = $1,010,000 + $136,000 + $10,000 + $100,000 = **$1,256,000**. **The target fell 49% in verification**, from about $950k. The original floor of “~$400k” was an addition

* Suggested formulation: check with a health worker before sharing.

error (its parts summed to $498.5k). Surge nights were cut from 20–45 to 10–40 because HSH already pays for its own activations (about ten nights last winter), so only nights beyond the city's triggers are fundable. Kits were resized from 3,400 citywide to 500–1,000 local. The SF Foundation money cannot be earmarked as five regrants. **Scope warning:** the surge beds are a citywide placement, since Sanctuary is in SoMa (94103), not this ZIP. The companion San Francisco city briefing carries the same line, so do not count it twice.

RECIPIENT	USE	AMOUNT (TARGET)	MECHANISM	BASIS	VERDICT
Episcopal Community Services (2024 revenue $67.7M; runs Sanctuary under a $40.9M HSH contract)	Action 1: storm-night surge beds and 24/7 staffing on nights beyond HSH's triggers	$378,750 (was $568k)	Standby contract paid per activation night; void if HSH expands its own surge	150 beds × 20 nights × $126.25 (SF Controller FY23 average, not marginal cost). Night count assumed from last winter's ~10	PARTIALLY CONFIRMED
SF Foundation Rapid Response Fund (pooled; $3k–20k grants)	Action 3: building operators' outage rosters and wellness support	$60,000 (was $100k)	Unearmarked gift to the pool, or direct grants to building operators	Fund is live with rolling applications. NERT is a Fire Department program and cannot be a grantee	PARTIALLY CONFIRMED
An outreach group that actually distributes supplies (not the Coalition on Homelessness, which is mainly advocacy)	Action 1: wet-weather kits (tarp, poncho, wool blanket, socks)	$40,000 (was $272k)	Pre-positioned stock	500–1,000 kits; the $40 kit cost is an assumption. SFHOT already hands out blankets and ponchos on activation	PARTIALLY CONFIRMED
Trusted outreach groups	Action 2: RV-listed CO alarms	$10,000	Pre-positioned stock	Unit cost and vehicle count assumed; no cost-effectiveness study	PARTIALLY CONFIRMED
Flood barriers	—	$0	Not funded	SFPUC already pays 100% of eligible costs (action 7)	ALREADY FUNDED

The funder's one thing: buy an option rather than spending now. The excess deaths here are few, and they cluster on specific storm nights among older people sleeping outside. A pre-signed standby contract that pays only when the weather trigger fires saves the most lives per

* Suggested formulation: check with a health worker before sharing.

dollar. The cheapest action in this briefing costs nothing: a supervisor's office asking HSH to publish its trigger and HSOC to pause clearings while shelter is open.

DO NOT DO THIS — CONTRAINDICATED

Advice that circulates for this hazard and would cost lives, money, or trust here.

Buy parametric household flood cover for this winter

CONTRAINDICATED

No parametric flood product exists for San Francisco households. The model one lens cited, New York's 2023 pilot (Swiss Re, Guy Carpenter, and ICEYE, paying through the Center for NYC Neighborhoods), triggered on satellite-measured flood extent, not rainfall, and ran for twelve months. The proposed trigger, 4.0 inches in 24 hours at the downtown gauge, would be a new design with high basis risk: it would miss king-tide flooding and pay on rain that floods nobody. Placing it needs a payer, a reinsurer, and a regulator, which cannot happen before the peak. Effort spent on it now is effort not spent on action 1.

Instead: property-level protection paid before the flood (action 7), and consider parametric cover for the 2027–28 season. California analogues to study are the Isleton household pilot and Fremont's municipal policy (Floodbase).

Clear encampments or tow lived-in vehicles in the days around a storm

DO NOT DO

Advocates report that San Francisco cleared camps days before the January 2023 atmospheric rivers and threw away survival gear, and KQED documented tent seizures before storms in November 2024. During one SoMa clearing in the rain, the city had ten beds to offer and people turned them down because they distrusted the city. A clearing on a storm day removes the tarp and sleeping bag that are a person's only protection, and scatters people away from the outreach workers who know where they are.

* Suggested formulation: check with a health worker before sharing.

Instead: a written pause on enforcement whenever the Cold & Wet protocol is active, with outreach carrying same-day bed counts (action 1).

Tell people a winter shutoff moratorium, or Medical Baseline, will protect them *

DO NOT DO

The CPUC's weather moratorium applies only when the forecast is below 32°F, which San Francisco rarely reaches, and Medical Baseline gives advance notice, not priority restoration. Advice built on either leaves a household believing it is protected when it is not.

Instead: apply for REACH and Match My Payment now (action 8), and plan backup power or oxygen for any power-dependent resident (action 3).

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Jan 1982	Bay Area storm; ENSO-neutral (ONI −0.1)	33 Bay Area deaths, 25 from landslides, 10 at Love Creek (Santa Cruz County)	USGS landslide warning research
1982–83	Very strong El Niño (ONI 2.14); SF season 38.17 in	Storm surge and high tides together in the Bay; record water level widely cited but not verified here	—
5 Nov 1994	SF's wettest day, 5.54 in; weak El Niño (ONI ~0.9)	Not found	—
1997–98	Very strong El Niño; SF season 47.19 in; wettest February, 14.88 in	No SF deaths found; statewide figures unverified	—
2015–16	Very strong El Niño (ONI 2.59)	SF about 98% of normal rain; the city ran sandbag pop-ups and opened expanded shelter	—

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
31 Dec 2022	SF's second-wettest day, 5.46 in; La Niña	EOC opened late; 17th & Folsom flooded; Vehicle Triage Center entrance flooded; 117 home and 17 business damage reports	17th & Folsom relief (Water Board order R2-2021-0021) finishing late 2027
Jan 2023	Atmospheric-river series	297 shelter beds open against 4,000+ unsheltered; clearings continued	No published after-action report found
21 Mar 2023	Wind storm	Two killed by falling trees (Post St; Lincoln Way)	—
4 Feb 2024	Wind storm	PG&E 1.1M customers out at peak across its territory	—
20 Dec 2025	Substation fire outage	~130,000 SF customers out; DAS ran wellness checks	—

The recurrence nobody has noticed: San Francisco's worst winter storms have not tracked El Niño strength. The deadliest (1982) came in a neutral year and the second-wettest day (2022) in a La Niña, while the strongest El Niño on record (2015–16) was near normal. The flooding follows the buried Mission Creek channel, at 17th and Folsom and along Mission Bay, in any ENSO phase.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **“Don't mention El Niño; say atmospheric rivers come in any winter”** (the Skeptic) versus **NOAA's 50–60% odds of a wetter-than-normal coast during a possibly record El Niño** (verification). Resolved as: say a wet winter is likely, not certain, and prepare as for any winter.
- **Neighbourhood sandbag pop-ups** (the Historian) versus **“sandbag yards reach nobody at risk”** (the Skeptic). Resolved as delivery plus pop-ups (action 6). What would settle it: a count of how many bags reached addresses with a backup history, which the city has never published.
- **Storm-night overdose as the hidden death toll** (Public health) versus **no evidence that storms raise overdose** (verification). Kept as an add-on, not a headline. What would settle it: linking Medical Examiner deaths among unhoused people to storm dates.

* Suggested formulation: check with a health worker before sharing.

- **Hypothermia deaths in San Francisco are near zero** (one undated count of two) versus **San Francisco among the highest hypothermia death rates of California's large counties** (KFF Health News). Unresolved; the Medical Examiner's exposure-death counts for recent years were not retrieved.

WHAT NOBODY COVERED — BLIND SPOT

Crowds and hospital access. No lens examined an atmospheric river on a Chase Center event night, with up to 18,000 people leaving into flooded streets and a disrupted T-line, or the routes into UCSF Benioff Children's Hospital if Third, 16th, or Mariposa Streets flood. And no lens followed the children: this ZIP is family-heavy, and the strongest outage-CO evidence (*Pediatrics* 2025) is about children. Both are cheap to check with the Warriors' event operations and UCSF emergency management, and neither has been verified.

HOW TO READ THIS — METHOD

The panel is author-built. Seven simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people or more. They are not neighbourhood figures. The El Niño projection covers *heat only*; flood, cold, wind, carbon monoxide, and overdose are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED — METHODOLOGY

- **Scope.** The request named “zip code 94158 in California”. It is treated as the USPS ZIP: southern Mission Bay plus parts of Dogpatch and east Potrero Hill. Climate Impact Lab impact region, as the report records it: San Francisco County, CA, *inferred* rather than resolved on the Impact Map; the city and county are coterminous (~830,000 people). Most statistics are for Supervisorial Districts 6 and 10 or the whole city, and are labelled as such. A companion briefing covers the City of San Francisco as a whole.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February

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2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Heat is not this place's winter hazard, so the historical record carries the hazard instead.

- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias: six lenses converged on storm-night shelter, but their evidence is local (San Francisco's own 2023 and 2024 record), so the convergence stands; the effect on deaths, however, is unmeasured anywhere, and the report says so. Contradictions were kept. Actions were ranked by expected lives saved per unit of effort before the peak, given payers or marked as within existing budgets, and reconciled against the funding placements.
- **Verification.** Seven verifier agents checked 13 recommendations, the historical record, the ENSO messaging, and the funding placements on evidence, transferability, and feasibility. Financial actions were also checked for availability, a named payer, a trigger, and whether they already exist. Verifiers queried the HHS emPOWER REST service, the NOAA CPC ONI table and ENSO discussion, NOAA tide predictions for station 9414290, the DataSF 100-Year Storm layer, and IRS filings on ProPublica directly. The envelope was re-summed.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Queried by one verifier: the nearest entries were a cold-weather alert program (Canada) and heat advisories (US), with nothing on outage welfare checks, CO alarms, storm shelter, or houseboats. Two other verifiers found the page rendered empty without its interface. Actions are tagged “queried, no precedent”, meaning no matching entry was found, not that the action is unprecedented.
- **Health wording.** Passages where a wording error could hurt someone (carbon monoxide, medical equipment in an outage, and backup power or oxygen) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so. Assumed unit costs ($40 kits, $50 alarms) and night counts are labelled as assumptions.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. The *Chronicle*'s 2022 sandbag reporting was paywalled. The ~138 NFIP policies citywide and the $364.3M HSH shelter line were not verified and are not used. The Medical Examiner's recent exposure-death counts, the 2026–27 HSH winter plan, DEM's current

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activation trigger, and the December 2024 update of the 100-Year Storm map were not retrieved. The Jan 1983 Bay water-level record was not verified.

References · verified by Phase 5 unless marked. NOAA CPC ENSO discussion, 8 Oct 2026 (verified) · NOAA CPC long-lead outlook (verified) · NOAA CPC ONI table (verified) · SF seasonal rainfall (verified) · SF Standard, wettest days (verified) · USGS PP 1434, January 1982 storm (verified) · HSH inclement weather protocol, Dec 2025 (verified) · HSH inclement weather protocol, Feb 2026 (verified) · Interfaith Winter Shelter (verified) · SF Controller shelter assessment, Dec 2024 (verified) · SF Standard, January 2023 shelter gap (verified) · KQED, tent seizures before storms (verified) · SF Standard, SoMa clearing in the rain (lens) · 2024 Point-in-Time Count (corrected) · 2026 Point-in-Time Count, preliminary (verified) · Pier 68 RV storage pause (verified) · KQED, RV parking ban (lens) · Toronto, cold injuries among unhoused people (lens) · Northrop et al., *Pediatrics* 2025 (corrected) · KTVU, Sea Ranch generator deaths (verified) · CA Health & Safety Code §17926 (verified) · HHS emPOWER REST service (queried 9 Oct 2026) · DAS Commission minutes, Jan 2026 (verified) · Mercy Housing, 1180 Fourth St (corrected) · Anderson & Bell 2012, NYC blackout mortality (verified) · SF Standard, February 2024 outage (verified) · SF Fire NERT (corrected) · *JAMA Netw Open* 2022, SF homeless deaths (corrected) · OCME overdose report, Sep 2026 (verified) · Walley et al., *BMJ* 2013 (verified; not transferable) · Cold and overdose, *Epidemiology* 2019 (verified) · SF naloxone doses distributed (verified) · KFF Health News, California hypothermia deaths (corrected) · L.A. Taco, hypothermia counts (contested) · SF Public Press, 2022–23 emergency response (verified) · SF Chronicle, NWS flood watch (verified) · Cal OES, California Disaster Assistance Act (verified) · SF Public Works sandbags (verified) · SF Public Works, 2015–16 sandbag season (corrected) · SF Standard, 17th & Folsom barriers (verified) · SFPUC Floodwater Grant terms, Nov 2025 (verified) · DataSF, 100-Year Storm Flood Risk Map (2022) (queried) · Wawona stormwater project (verified; not this grant) · SFPUC Folsom Area Stormwater project (verified) · PG&E REACH (verified) · PG&E Match My Payment (verified) · PG&E $50M 2026 pledge (verified) · TURN, CPUC disconnection rule (verified) · Swiss Re, NYC parametric flood pilot (verified) · Floodbase, Fremont municipal policy (verified) · SF 2023 small-business flood grants (verified; one-off) · NOAA tide predictions, station 9414290 (queried) · Baykeeper, Mission Creek houseboats (verified) · Port of SF story map (lens) · NBC Bay Area, March 2023 tree deaths (verified) · SF Public Works, street trees (Prop E) (verified) · Episcopal Community Services, IRS filings (verified) · HSH contract, ECS Sanctuary (verified) · SF Foundation Rapid Response Fund (verified) · Statistical Atlas, ZIP 94158 (lens) · SF floodplain administration (verified)
Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory
Generated 9 October 2026 for ZIP 94158, San Francisco, California. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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